LAB 17.1

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Task 1: Customer Data Cleaning

Prompt

Clean customer dataset using Pandas by handling missing values, duplicates, and encoding.

Code

```
import pandas as pd

df = pd.DataFrame({
    'age': [20, None, 25],
    'income': [50000, 60000, None],
    'city': ['Chennai', 'DELHI', 'Mumbai'],
    'gender': ['Male', 'Female', 'Male']
})

df.fillna(df.mean(numeric_only=True), inplace=True)
df.drop_duplicates(inplace=True)

df['city'] = df['city'].str.lower()

df['gender'] = df['gender'].map({'Male':0, 'Female':1})
df['city'] = df['city'].astype('category').cat.codes

print(df)
```

Output

```
   age   income  city  gender
0 20.0 50000.0     0       0
1 22.5 60000.0     1       1
2 25.0 55000.0     2       0
```

Task 2: Sales Preprocessing

Prompt

Preprocess sales data by converting dates, creating features, and normalizing values.

Code

```
import pandas as pd

df = pd.DataFrame({
   'date': ['2024-01-01','2024-02-01'],
   'sales': [1000, 2000]
})

df['date'] = pd.to_datetime(df['date'])
df['month'] = df['date'].dt.month
df['year'] = df['date'].dt.year

df['sales'] = (df['sales'] - df['sales'].min()) / (df['sales'].max() - df['sales'].min())

print(df)
```

Output

```
       date  sales  month  year
0 2024-01-01    0.0      1  2024
1 2024-02-01    1.0      2  2024
```

Task 3: Healthcare Data

Prompt

Clean and preprocess healthcare dataset with scaling and train-test split.

Code

```
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
import pandas as pd

df = pd.DataFrame({
   'bp': [120, None, 140],
   'chol': [200, 220, None],
   'gender': ['M','F','M']
})

df.fillna(df.mean(numeric_only=True), inplace=True)
```

```
df['gender'] = df['gender'].map({'M':0,'F':1})

scaler = StandardScaler()
df[['bp','chol']] = scaler.fit_transform(df[['bp','chol']])

train, test = train_test_split(df, test_size=0.2)

print(train)
```

Task 4: Employee Salary

Prompt

Preprocess employee salary dataset by removing outliers and encoding categories.

Code

```
import pandas as pd

df = pd.DataFrame({
   'salary':[30000,40000,1000000],
   'department':['HR','IT','IT'],
   'location':['Hyd','Chennai','Delhi']
})

df['department'] = df['department'].str.lower()

df = df[df['salary'] < 500000]

df['department'] = df['department'].astype('category').cat.codes
df['location'] = df['location'].astype('category').cat.codes

print(df)
```

Output

```
   salary  department  location
0  30000         0         1
1  40000         1         0
```